A Team-Based Win Probability Model for NBA Basketball Games: Extending Stern (1994)

Abstract

This paper extends Stern's (1994) Brownian motion model for NBA win probability by incorporating pregame betting spreads as a game-specific estimate of team advantage. While Stern's original model treated home court advantage as a fixed constant, this paper replaces it with a variable derived from pregame odds. Using a merged dataset of 18,078 NBA regular-season games from 2008 to 2022, a maximum likelihood algorithm was used to estimate the variability parameter. The updated model outperformed the original across all five out-of-sample test seasons, achieving a mean Brier score improvement of 3.00%, corresponding to approximately 35 games per season predicted differently. Additionally, the assumption of constant variability throughout a game was tested using play-by-play data and found to be largely valid. Building on the model, a strategic analysis was developed to quantify the leverage of scoring at different points in a game. Results indicate that in tied games, scoring is most valuable in the final minutes of the fourth quarter, while teams leading by ten points at halftime benefit most from extending their lead at the start of the third quarter. These findings offer practical, data-driven guidance for coaches on player rotation and minute management.

Key Words: Maximum likelihood estimation, Brownian motion, Pregame spread, Brier score, Coaching strategy, Point leverage

Introduction

Watching basketball always fills me with excitement, wondering whether they will score in the next play, who will make the shot, and ultimately, who will win the game. Growing up, basketball has always been a part of my life, and every time it came down to a close game, I would start to feel butterflies in my stomach. I still remember being at a game between ASF and TEC CSF when ASF came back from 19 points in the fourth quarter and sent it to overtime; my heart was racing. But why? Why does it feel like scoring in the final moments is 'worth more' than a basket at any other time in the game? And what do statistics say about when the star player should rest? To answer these questions, I extended Stern's (1994) model. This paper explains how Stern used Markov chains and Brownian motion and then improves the model by using the pregame spread to increase accuracy and develop a strategic analysis for coaches. The pregame spread is a number set by oddsmakers that represents the expected point margin of victory for the favored team.

Related Work

When it comes to win probability models of basketball games, there are two primary methodologies used: a continuous-time model and a discrete possession-based model. In this paper, the continuous-time model is used, with variables adjusted using techniques inspired by more recent work.

The foundational continuous-time model was developed by Stern (1994). This model used the current lead and time as inputs to determine the win probability and treated the change in the score margin as a continuous random process. The model is characterized by two parameters: homecourt advantage (μ) and the variability (σ). These two parameters were fixed to

the dataset and relied on two simplifying assumptions. Choosing a fixed advantage for the home team disregards the games where away teams are better and have an initial advantage. Moreover, assuming stable variability eliminates the possibility that it changes as the game progresses. Despite these limitations, the model predicts basketball outcomes well and has been the foundation for future work in sports win probability modeling.

Subsequent work by Polson and Stern (2015) expanded on this model by incorporating information from betting markets. Instead of treating the parameters as fixed, they demonstrated that market data can be used to infer the model parameters. In their model, the point spread tells you the expected margin of victory, and the betting odds help determine how much uncertainty there is in the game. Instead of trying to build a better prediction model, their main goal was to show that betting markets already contain useful statistical information about the model's parameters. They focused more on interpreting what the market reveals rather than improving prediction accuracy or testing different versions of the model.

Similarly, Strumbelj & Vracar (2012) researched betting markets to estimate win probabilities and team strength. Instead of modeling the internal mechanics of a game, they treat betting markets as highly informative and statistically efficient. Their method refines market-implied probabilities to produce calibrated predictions, showing that betting odds can serve as a strong baseline for forecasting outcomes.

On the other hand, Shirley (2007) developed a model to use a Markov chain, where each state represented a possession configuration defined by team control and recent scoring events. Transition probabilities between states were estimated from team statistics. This allowed them to simulate games against average teams. It explained how the game works play-by-play, including

possession changes and types of scoring. However, it used many parameters and relied on estimated probabilities; it was also harder to compute than continuous-time models.

Even though these models have improved over time, not many researchers have carefully tested the assumptions behind continuous-time models. In particular, most studies assume that key parameters like the expected scoring advantage and variability (how much randomness there is) stay constant throughout a game.

This study looks directly at whether those assumptions are actually true. It allows the pregame advantage parameter to change from game to game based on pregame expectations from betting markets, and it tests whether variability changes during different parts of a game.

The main contribution of this study is about improving how the model is calibrated, not creating a completely new type of model. The basic mathematical framework is still the same as Stern's original model, but the parameters are reexamined, tested, and updated using real data. In that way, this work builds on existing research by refining and validating the model rather than replacing it.

Mathematical Foundation: Markov Chains

At the basis of this win probability model is the idea of Markov chains. These are chains of events whose probability of moving on to the next event depends solely on the current state. In the case of NBA games, Markov chains were adapted with the following criteria: the states or events are point differentials, where the lead is calculated from the home team (i.e., a -6 lead refers to the home team losing by 6 in that moment), and the transitions are the change in possessions. To put it in simpler terms, a basketball game moves between states, which refer to the score of the game, and every time there is a change in possession, there is a set of

probabilities for the team with the ball to score or not score. The most important part of this explanation is to remember that the probability of the lead varying with a change in possession depends only on the current lead, not past events.

To dive a little bit into the math behind why Markov chains work to represent basketball games, it is imperative to talk about stochastic matrices. Stochastic matrices refer to those that contain the probabilities for an event to change into the next when only remembering about the event they are currently in. This is much like Markov chains, which are exactly what they are used for; however, what is interesting about these matrices is some of the properties they have. The eigenvalue $\lambda = 1$ corresponds to a steady-state vector, a probability distribution that remains unchanged with further transitions. In basketball, this represents the outcome: once the game ends (the steady-state), the probabilities become fixed at either 100% win or 100% loss for each team. There is a limitation to using Markov chains to explain the win probability of basketball games; the large number of discrete states and complex transition probabilities make the chain complicated to calculate.

The Brownian Motion Model

The discrete Markov chain approach, while theoretically sound, requires estimating transition probabilities for hundreds of possible score changes at each point in the game. Stern (1994) proposed modeling the score differential as a Brownian motion process. A Brownian motion is a type of Markov chain, distinguished by its continuity. Markov chains are usually limited to discrete states, where the transitions between them occur in ‘jumps’.

On the other hand, Brownian motions follow a continuous path between states. Brownian motion has a key property: the change in score over any time interval follows a normal

distribution with mean and variance proportional to the interval length. This makes it well-suited for basketball, where scoring happens relatively continuously and the score differential at any moment can be approximated as normally distributed around an expected value.

Returning to Stern's research, he created a formula to compute the probability that the home team wins, which was the following: $P(win|e,t) = \frac{\phi(e+(1-t)\mu)}{\sqrt{(1-t)\sigma^2}}$ where e is the current lead, t is the time going from 0 to 1, where t = 0 corresponds to tipoff and t = 1 refers to the end of the game. For reference, the end of each quarter corresponds to t = 0.25, 0.50, 0.75, and 1, respectively. In Stern's original model, μ was a fixed constant representing the average home court advantage across all games. In this paper, μ is replaced with a game-specific value derived from the pregame betting spread, capturing the relative strength between the two teams playing. Lastly, σ represents the standard deviation of the dataset; this value was calculated using a maximum likelihood algorithm.

Building the Model

For the analysis, a dataset that included the score differentials at each quarter and the pregame spread for every game was needed. Since I wasn't able to find one, I merged two datasets based on the home team, the away team, and the final score of the game.

The analysis uses data from the NBA Official Website, obtained through Kaggle, which contains 71,267 games from 1949 to 2024. This dataset included columns for the final score at each quarter for every game, with which I was able to calculate the point differential at those times. The dataset was filtered to only regular-season NBA games, removing All-Star and Exhibition games.

For the pregame betting spread, data from Sportsbook Review, obtained through Kaggle, was used. It contained roughly 18,500 games from 2008 to 2023. Since each game appeared twice, once from each team's perspective, only the home team entries were kept. I then changed the sign on the pregame spread to make the home team advantage positive. This column served as the per-game estimate of μ, replacing the fixed home court advantage used in Stern's original model.

To merge the two datasets, first, I had to make sure that the columns I was going to match up the games with were in the correct formatting. This required mapping full city names to abbreviated team names, for example, LA Lakers to Lakers. Since the 2023 season was incomplete in the pregame spread dataset, that season was excluded from the training dataset and then used to test the accuracy of the model. Then, after the two datasets were merged, I was left with 18,078 games from 2008 to 2022.

To find the appropriate value of σ (the variability) for my dataset, I built a maximum likelihood algorithm. To begin, I input an initial value of $\sigma = 15$, which is what Stern used in his analysis, then the algorithm used the formula with μ set to the pregame spread for each game to calculate the probability of the home team winning for each game. To combine all those individual probabilities, the optimizer would take the log of each probability and then add them. This was an important step because, since all the probabilities were so small, just multiplying them together would result in a loss of precision, and it would round out to zero. It then adjusted the initial guesses for the value and repeated the process until it achieved the highest log-likelihood possible. Using this system, I obtained the following value $\sigma = 16.74$ points. This value is extremely similar to the one Stern found ($\sigma = 15.82$), validating my result.

The Brier score measures the average squared difference between the predicted probability and the observed outcome, with lower values indicating better predictions. The Brier score was calculated for each of the five seasons using both Stern's original fixed μ model and the variable μ model developed in this paper. These values can be found in Table 1. The updated model outperformed the original model in all 5 seasons. These improvements ranged from 2.42% to 4.07%, with a mean improvement of 3.00% and a standard deviation of 0.74%. A low standard deviation demonstrates a consistent improvement in the model. To contextualize this, applying a 50% decision threshold to classify each game as a home win or loss, approximately 35 games per season changed their classification at 0.5, reflecting meaningful probability shifts.

[Insert Table 1 HERE]

Validating the Constant Sigma Assumption

When thinking about how basketball games usually pan out, it makes sense that the variability of the score changes as the game progresses. This is because, in general, teams are more willing to take risks as time winds down. For example, fouling intentionally or a full-court press can create big leads or make blowouts close games.

To test if variability changes throughout the game, I used data from the CDN NBA data repository. The dataset included every regular-season game from 2020 to 2026, totalling 3.6 million rows. The dataset contained the home team, the away team, the score for each team, and the time left on the clock. This means I could calculate the lead at any point during the game. With this data, I divided the game into 8, 6-minute segments. I then repeated the same maximum likelihood algorithm described before at each segment and was able to create a graph with the

variability at each point, seen in Figure 1. The objective with that graph was to use the line of best fit as σ instead of a fixed number.

[Insert Figure 1 HERE]

In early game windows, the likelihood is flatter, reducing identifiability; we therefore excluded the first segment. The second segment of the game, which is the earliest included in the analysis, has the highest variability. This makes sense because at the beginning of the game, teams can go on short runs and neither team has settled into the game yet. Then, for the entirety of the game, the variability changes very minimally, contrary to my hypothesis. I thought that as the game progressed, teams would be taking more risks and therefore increase the variability; however, the graph clearly shows a downward trend. I think this can be attributed to the fact that a lot of games aren't close, and as time goes on, teams grow the leads, and the game is close to over before getting to the fourth quarter. Interestingly, in the last segment of the analysis, there is a clear spike in variability that can be explained by close games. This point follows my initial guess that in the final stages of the game, teams would be more hectic and increase the variability.

To quantify the improvement, I calculated the Brier score with the constant σ and with the variable σ. The results show a 0.51% improvement in Brier score from 0.1333 to 0.1326. I found that the change was not large enough to meaningfully improve the performance. This result helps confirm Stern's assumption that σ it should be constant, which was never explicitly tested.

Strategic Analysis

Using the win probability model, I began by looking at the lead required to have a 95% chance of winning at every point in the game for different pregame spreads. The 95% threshold was chosen because it represents a strong but realistic standard. A team with 95% win probability has the game well in hand, while 99% would require leads so large they rarely occur in practice. To construct this graph, the model was inverted to find the lead needed to get the specific win probability for each pre-game spread and time. Each pregame spread had a sample size of roughly 3,000 to 6,600 games, which ensures that the results are not driven by a small sample. At first, looking at Figure 2, nothing really stood out as it makes sense that as time goes on, the amount of points needed to maintain a high win percentage goes down at a faster rate. However, there are still some interesting things the graph shows, for example, the red line referring to games where teams have a 15-point pregame spread is interesting because of the low amount of points they need to maintain a very high chance of winning. In fact, they never even need to have a 15-point lead to have a 95% chance of winning. This graph also shows how, for most lines, teams can rapidly lose leads at the end of the game and continue to have a high win percentage. This raises the question of how valuable points are at specific times of the game. And at what time do points matter most? With these answers, coaches would be able to know when they can rest their starters and when scoring is most important.

[Insert Figure 2 HERE]

To answer these questions, I built a couple of graphs that show the increase in relative win percentage as the game progresses. First, I made one that showed the “value” of a point that

took the game from tied to up 1 for the home team, seen in Figure 3. The model calculated the win probability at each time with the lead being 0 and then with the lead being 1, and then it calculated the relative increase in win probability. From this graph, I concluded that in a tied game, the most important moment is the second half of the 4th quarter. That is where the huge spike in relative win percentage happens, and that means that it is crucial to take the lead at that time.

[Insert Figure 3 HERE]

With these findings, the logical next step was to experiment with different leads. I tried expanding the lead from 5 to 6 and from 10 to 11, as seen in Figure 4. This revealed powerful insight into how coaches should be managing their best players' minutes. When teams increase the lead from 5 to 6 points, regardless of the pregame spread, there is a peak at about the beginning of the 4th quarter and then a sudden drop. This signifies that in a game where a team is roughly up 5, the most important time to have your starters in is at the beginning of the 4th quarter, because that way you can increase the lead and the game will just about be over.

Moving on to the graph for the 10 to 11 point lead, an even more striking discovery exists. The peak for the increase in relative win percentage occurs roughly at the beginning of the second half. This means that teams going into halftime roughly up 10 should be using their starters at the beginning of the second half to increase the lead, and then the game will be over, so the starters can rest.

[Insert Figure 4 HERE]

To summarize the insights from the point leverage graphs, the following conclusions can be drawn. If a game is tied, it is ideal to rest your starters in the third quarter so they are ready for the most important part of the game, the second half of the 4th quarter. If a team is up roughly 5, the most important part of the game to increase their lead is the beginning of the 4th quarter. Lastly, when teams enter halftime roughly up 10, they should be aiming to increase the lead in the third quarter. The opportunity costs of resting starters at strategic times during the game can be quantified. In a tied game, scoring in the 4th quarter is 2.07x more valuable than in the third quarter. However, when up 10, scoring in the third quarter is 2x more valuable than in the 4th quarter. It is also important to note that strategies differ depending on the pregame spread. In general, the higher the pregame spread is in favor of the team, the later in the game the crucial moment of scoring happens. In other words, if you are expected to win by a larger margin, you have a bigger safety net that keeps the win probability high for a longer time. This means that the moment when scoring becomes most critical occurs later in the game.

Finally, I wanted to know the “point of no return”, or in other words, when the lead was too big to overcome, and the game was practically over. To do this, the win probability was calculated at every time point and plotted against the 95% threshold line, showing when and whether each combination crosses into safe territory, as seen in Figure 5. The results are pretty intuitive as they show that small leads are very easy to overcome and comebacks do happen all the time. It makes sense that the bigger the lead, the earlier in the game teams are able to cross the line of safe win percentage. However, it does shed light on an interesting behavior of the model: when teams with a pregame spread of 15 are in trouble and only up 2, the model loses confidence in the team very quickly. This is seen by the sharp decline in win probability, although at around the 4th quarter, the team's win probability begins to rise again. It is certainly

interesting that the model has such high expectations for a team that was supposed to win the game by 15 and loses it quickly when it is only up 2.

[Insert Figure 5 HERE]

Limitations

Even with these findings, the model has limitations. Most obviously, some teams are just better than others; they can have better players and team chemistry. This is partially addressed by the addition of the pregame spread, but some things, like injuries and in-game momentum shifts, aren't taken into account. Then, specifically for late-game situations, there are a lot more variables involved in who wins the game, like possession of the ball, foul bonus, personal fouls, and position of the ball, among others. To put it simply, basketball is a complicated game with many variables, making it nearly impossible to account for all. Despite these limitations, the lead and time remaining are the most critical factors, and the model provides a reasonable approximation of win probability.

Even if addressing all variables is unreasonable, there are a few worth trying to include in future models. First, a team-specific sigma derived from team statistics from the previous season would encompass more accurately how consistent teams are. Then, it would be interesting to see how much a model improves if it considers possession of the ball in late-game situations. Lastly, when talking about when to rest starters and how to manage their fatigue, the idea of modeling player fatigue came to mind. This could help coaches know how tired their players are and when they need to rest more accurately.

Conclusion

In conclusion, points really do matter more in the final stretch of games. This research started with Markov chains and how they can predict the next step, with only information from the current state. However, scoring in basketball games happens in a continuous way and not really in jumps. This is why Brownian motions were used, as they are continuous within a given interval. I was able to adjust Stern's equation to include the pregame spread of games μ, obtaining a model I could use to analyze strategic coaching advice. Alongside adding to Stern's model, I managed to validate the assumption that the variability should be constant. I did this by graphing the variability at segments throughout the game and found that they remained roughly constant. The model showed an average improvement in Brier score of 3.00% over Stern's original, corresponding to approximately 35 games per season predicted differently. With this model, multiple graphs were created to analyze strategic coaching insight. The analysis revealed a counterintuitive insight that if you are tied at halftime, you should be resting your starters for the more crucial part of the game in the fourth quarter, and if you are up 10, you should be playing the starters to increase the lead. Although there are some limitations to building win probability models for basketball, accuracy is not severely compromised if the correct variables are chosen. With my model, I was able to calculate the probability of ASF coming back and sending the game to overtime. TEC CSF had a 99.4% chance of winning at the beginning of the fourth quarter, meaning it was virtually impossible, but ASF beat the odds.

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